Exercise 1: Feature Selection Using Information Gain
Dataset:
Use the Titanic dataset (available on Kaggle or seaborn library).
Objective:
Identify the most important categorical features for predicting survival using information gain.
Instructions:

1. Dataset Overview:
   - Target: Survived (0 = No, 1 = Yes).
   - Features: Use categorical features only:
     - Pclass (Passenger class: 1st, 2nd, 3rd).
     - Sex (Male, Female).
     - Embarked (Port of Embarkation: C, Q, S).
2. **Task**:
   - Calculate the information gain for each categorical feature (e.g., Sex, Pclass, Embarked) with respect to Survived.
   - Rank the features based on their contribution to predicting survival.
3. **Hints for Implementation:**
   - Use a contingency table to manually calculate information gain for one feature (e.g., Sex) if working on paper.
   - For automated methods, use libraries like sklearn.feature_selection.mutual_info_classif after encoding categorical features with one-hot or label encoding.
4. Expected Output:
   - A ranked list of features by their information gain.
   - Discuss which features are the most important and why.

Challenge:
Visualize the relationship between the most important feature and the target using a stacked bar chart or count plot.

Exercise 2: Feature Extraction Using PCA
Dataset:
Use the Iris dataset.
Objective:
Reduce the dimensionality of the Iris dataset using PCA and analyze the impact on data structure.
Instructions:

1. Dataset Overview:
   - Target: species (Setosa, Versicolor, Virginica).
   - Features: sepal length, sepal width, petal length, and petal width (all numerical).
2. **Task**:
   - Standardize the features using StandardScaler.
   - Apply PCA and reduce the dataset to 2 components.
   - Visualize the data in the 2D PCA space.
3. **Hints for Implementation:**
   - Class to use: sklearn.decomposition.PCA.
   - Use explained_variance_ratio_ to understand how much variance is retained by the first two components.
   - Plot the data points in the PCA-reduced space, color-coded by species.
4. **Expected Output:**
   - A scatter plot showing clusters of flower species in the PCA space.
   - Analyze whether flowers of the same species cluster together in the reduced space.

Challenge:
Explain the variance retained by the PCA components (e.g., how much variance the first two components capture) and discuss the trade-off between reducing dimensionality and preserving information.

Impact Visualization:

1. Before and After Feature Engineering:
   - Train a simple classification model (e.g., Logistic Regression) on:
     - The original 4 features.
     - The 2 PCA-transformed components.
   - Compare performance using metrics like accuracy.
2. **Visualization**:
   - A scatter plot showing the PCA-reduced space, color-coded by species.
   - A bar chart of the explained variance ratio for each PCA component.